SCALE100X.AI · PRODUCT TAKE-HOME

Task 1: Product Thinking

Franchise-in-a-Box OS

At a glance: the questions, the answers, the numbers

1a Prioritisation: what ships in the first 30 days

1b AI vs human split: is "76% AI" real?

1c Developer-ready spec: Conversation Takeover

Appendix: assumptions and decisions

15 October 2026

At a glance

What Task 1 asked

Part	The question, in plain words
1a	If only part of the platform can be built in the first 30 days, what goes first, and why?
1b	The brief says the platform runs on "76% AI, 24% human". Is that realistic?
1c	Take one feature the brief describes vaguely and write it up so a developer could build it without guessing.

The answers

1a: Build the part that earns revenue first. That means an AI WhatsApp sales agent that answers new franchise leads within minutes, plus a simple CRM where the sales team takes over and closes. The ROI calculator comes next, then brand onboarding. Why: Scale100x is paid when a franchisee signs, and the pilot brands are already onboarded, so the sales path can start on day 1. A checkpoint on day 5, using real sales data, confirms the order or changes it.

1b: Believable for workload, misleading as worded. Mapping all 37 tasks on the platform shows automation covers 68–87% of the work, so 76% is plausible. But only about 54% is genuinely AI; about 34% is ordinary software (templates, schedules, formulas). And people still make every high-value decision. The recommendation: replace the fixed label with a measured, audited figure: **"AI: X% of work | Humans: 100% of closes"**. Beyond the label, every workflow is judged end to end (data, integrations, approvals, failure paths, security, cost), and success means a business outcome, not just a task that ran.

1c: A developer-ready spec for "human takeover" of AI chats. The brief gives this one line. The spec defines who owns a conversation at every moment and what happens when two people grab it or nobody does (a 10-minute response rule with escalation). It makes sure the AI never replies after a person takes over, shows whether the agent is really sending (Active / Paused / Error), and covers WhatsApp's 24-hour messaging rule and opt-out, with 10 edge cases, 11 pass/fail tests, pilot metrics and a definition of done.

Headline numbers

Measure (per brand, per month, modelled)	Without AI	With AI
Hours of work	457	59
Brands one consultant can handle	0.35	2.7
Human hours per signed franchisee	~152	~20
Share of work time automated	—	87% (54% AI + 34% software)

These come from a model with stated estimates (lead volumes, minutes per task). A pilot checks them; the method is the point, not the exact percentages.

How the work was done

- Modules were ranked on five weighted criteria, each score with a written reason.
- The 37-task map lives in a live Google Sheet; every number in 1b comes from its formulas.
- WhatsApp platform rules were checked against Meta's own documentation.
- Every assumption and decision is labelled (A- and D- numbers) and listed in the appendix.
- The three parts fit together: the 1c spec is the core of the 1a plan, and it records the events used to measure 1b.

Terms used

Term	Meaning
Brand / franchisee	The brand sells franchises; a franchisee buys one and opens an outlet.
Lead	A prospective franchisee who has shown interest, usually through a Facebook or Instagram ad form ("Lead Ads").
CRM	The system the sales team uses to track leads and deals.
HITL	Human-in-the-loop: a person steps into a conversation the AI is handling.
ROI	Return on investment: what a franchisee can expect to earn back, and how fast.
SLA	A committed response time (here, a flag must be claimed within 10 business minutes).
BSP	A Meta-approved provider of WhatsApp Business API access.
Autonomy Rate	The proposed audited measure of how much work AI completes with no human edit.

Reading the rest: each part opens with a TL;DR box, then goes into the detail product and engineering reviewers need. **Online versions:** live Sheet · Google Doc · GitHub

1a. Prioritisation: what ships in the first 30 days

> **TL;DR.** First, ship the lead-to-close slice: the WhatsApp Sales Agent and a thin CRM, released together. Next comes a thin ROI Engine, so the agent has approved numbers to quote, then thin Onboarding. Collateral Studio waits. The order follows the money: Scale100x is paid when a franchisee signs (A1), and its 20+ existing brands (A21) mean the revenue path can start on day 1. The ranking is a hypothesis, not a fixed bet. Days 1–5 test it with discovery while the team does only work that every outcome needs, and a decision gate on day 5 says what changes if the evidence points elsewhere.

1. The lens

Scale100x's stated goal is to help brands open **100+ offline outlets a year**, and every one of those outlets is a signed franchisee. Scale100x earns a setup fee and retainer from brands, and a success fee when a franchisee signs (A1). The modules split into two groups:

- **Brand-side modules (Onboarding, ROI, Collateral)** earn the right to sell.
- **Franchisee-side modules (Sales Agent, CRM)** produce the signed agreements. They are also where most of the AI leverage sits: about 73% of all manual work per brand is in the Sales Agent module (1b).

Every module is ranked by one question: *does it get more qualified franchisees to a signed agreement, faster, across many brands at once?* A module doesn't rank higher just because it contains more AI.

2. Which bottleneck? Four hypotheses

The brief describes five modules but doesn't say which customer problem hurts most. This ranking rests on hypothesis H1, and discovery tests all four:

#	Hypothesis	What it would look like in the data	Module it would promote
H1	**Lead handling:** leads get slow replies or no follow-up	Long time to first reply; leads with no second touch; conversion falls with reply delay	Sales Agent + CRM (current #1)
H2	**Lead quality:** ads attract people who can't or won't invest	Many replies, few qualify; low-budget or wrong-city leads dominate	Qualification rules and lead-source analysis inside the Sales Agent
H3	**Investor confidence:** prospects stall at the numbers	Drop-off after the investment or payback conversation; repeated ROI questions	ROI Engine moves to #1 alongside the agent
H4	**Brand readiness:** brands lack the facts and documents to sell	Consultants can't answer basic questions; missing P&L or FAQs	Onboarding moves above ROI

H1 is the working bet, for two reasons:

- The money is earned at signing, and leads go cold within hours, not days.
- The 1b action map puts most of the manual work in the conversation itself.

That is a model of the work, not observed pain. It still has to be confirmed with funnel data.

3. How the modules are scored

Criterion	Weight	Question
Business impact	30%	Does it move franchisees closer to signing?
Customer pain	25%	Does it fix a frequent, costly problem?
30-day feasibility	20%	Can a useful, testable version ship in 30 days with this team (A7)?

Criterion	Weight	Question
Dependency leverage	15%	Do other modules need it?
Evidence confidence	10%	How much evidence backs the need today?

These weights are proposed decision weights, not measured results. The scores below (1–5) are pre-discovery judgments. Each comes with a written reason so it can be challenged, and all of them are re-scored after day 5.

Module (thin scope)	Impact	Pain	Feasibility	Leverage	Evidence	**Weighted**
HITL Sales CRM	4	4	4	4	3	**3.90**
AI Sales & Marketing Agent	5	4	3	3	3	**3.85**
Franchise Model & ROI Engine	3	3	4	4	2	**3.25**
Brand Onboarding	2	2	5	4	2	**2.90**
Collateral Studio	2	2	3	1	2	**2.05**

Reasons for the scores that matter most:

- **Sales Agent impact 5:** it is the step where a lead becomes a conversation.
- **Sales Agent feasibility 3:** Meta approvals are outside the team's control (§6).
- **CRM leverage 4:** the agent's takeover and audit both depend on its timeline.
- **ROI leverage 4:** its approved Model Card is the only source of numbers the agent may quote (A32).
- **Onboarding impact 2:** the pilot brands are already onboarded (A21).
- **Collateral leverage 1:** no other module needs it to work.

The CRM and the Agent score within 0.05 of each other, which is why they share rank 1 and ship as one slice.

4. Ranking (decision D-04, confirmed)

Team: 2 full-stack engineers, 1 designer and a PM, using AI tools on React + Supabase (A7). Pilot: 2 existing brands (A21).

Rank	Module	30-day scope (what ships)	Why now	Left out until later
1	AI Sales & Marketing Agent (thin)	WhatsApp AI Agent only: Lead Ads webhook → dedupe by phone number → consent → template greeting → qualification → answers from approved facts → HITL flag and takeover; basic live tracking (counts of leads, messages and conversations per brand)	Leads stop going cold because the reply arrives in minutes; this is where 73% of the work is	Voice and Email agents, Meta Ads optimisation, the full switchboard, live-tracking dashboards beyond basic counts
1	HITL Sales CRM (thin), shipped as one slice with the agent	Inbox, owner per lead, unified timeline, next action with follow-up reminders, takeover panel, deal stage up to Won / Lost	Without a human closing path the agent produces chats, not signed agreements	Live call intelligence, forecasting
3	ROI Engine (thin)	Franchise Model Card + 3-year ROI with editable, labelled assumptions, approved by a consultant and published as the agent's **approved facts**	Prospects' first serious question is about returns. The agent may only quote numbers a human has approved (A32).	Benchmarks (shown only when sourced, A33), AI insights
4	Brand Onboarding (thin)	Brand details form, file upload, required-field checks; a consultant enters data for the 2 pilot brands	Needed to add brand #3 onwards, but the pilot brands are already onboarded	AI Readiness Score, revenue-split donut

Rank	Module	30-day scope (what ships)	Why now	Left out until later
5	Collateral Studio	Deferred: the pilot uses each brand's existing deck plus a small set of approved WhatsApp templates	Existing brands already have decks; designers can make creatives with current tools	Everything

Why ROI ranks above Onboarding. A reasonable case can be made for the opposite order, because the agent needs approved brand information to answer well (H4). The two modules split that information differently:

- **Onboarding** holds brand details (name, category, cities). For the 2 pilot brands, a consultant can type these in.
- **ROI** holds the information prospects actually press on: investment, fee, royalty, payback. That is the part that has to be right and approved.

So ROI is the module that unblocks good answers. If discovery finds the pilot brands' basic facts are missing or inconsistent (H4), the two swap places.

Reuse before build: scheduling uses a calendar-booking link, and reminders use the CRM's own task list. The team doesn't build email or calendar tools in the first 30 days.

5. The dependency trap

The brief lists the modules in brand-journey order: Onboarding → ROI → Collateral → Sales → CRM. Building in that order looks logical, but it means the **first lead isn't handled until three brand-side modules have shipped**. At our capacity (A7) that falls outside the 30-day window, and the month would end with polished onboarding screens and no signed franchisees.

Reversing the order works because the pilot brands are **already onboarded** (A21). Their data can be entered by hand while the slice that earns the success fee goes live.

6. Day-1 critical path: WhatsApp access

The slice depends on outside approvals that the team can't speed up:

- WhatsApp Business API access through a BSP
- Meta business verification
- approval of the greeting and re-engagement **message templates**

Templates matter because a free-form message can only be sent within 24 hours of the prospect's last message. Outside that window, Meta returns error 131047 and only a template can be sent (Meta error codes).

These applications go in on day 1, in parallel with discovery and the build. No lead time is quoted here because none has been verified. The fallback if approval is late: run the pilot on the CRM with consultants replying by hand, and switch on the agent once access lands. The CRM work isn't wasted either way.

7. The 30-day roadmap

Days	Work	Exit gate
1–5: Discover, and start the no-regret work	**Discovery:** interview the pilot brands' sales consultants; pull whatever funnel data exists (lead volume, time to first reply, lead → call → signing); map the current lead lifecycle and where leads drop. **No-regret work** (needed under all four hypotheses): submit the BSP and template applications; build the CRM data model and the `ACTION_EVENT` log (1b §5); enter the pilot brands' data.	**Decision gate (day 5):** which hypothesis does the evidence support? Re-score §3, confirm or change the ranking, and record baselines for the §8 metrics.
6–12: Workflow foundation	Lead intake from Lead Ads, dedupe, owner and status, CRM timeline, follow-up reminders, access control, error handling	A test lead flows end to end with a human replying

Days	Work	Exit gate
13–21: AI assistance and handoffs	Agent qualification and answers grounded in approved facts; low-confidence and off-topic handling; flag → takeover (the 1c state machine); thin ROI Model Card published as approved facts	The 1c acceptance tests AC1–AC11 pass on a scripted test set
22–26: Pilot	Go live on the 2 pilot brands; run the first weekly audit (1b); fix defects; thin Onboarding form in parallel (it isn't on the pilot's critical path)	0 wrong answers in the audit; flags are being claimed
27–30: Measure and decide	Compare against the day-5 baselines; review the audit, adoption and time saved	Decide what comes next: add a channel, deepen the CRM, or promote ROI

What changes at the day-5 gate:

- **H1 confirmed:** keep the plan.
- **H2 (lead quality):** keep the slice, but move effort from answering to qualifying, with stricter qualification rules and reporting by lead source. A faster reply to the wrong lead doesn't help.
- **H3 (investor confidence):** pull the ROI Model Card into days 6–12 and make it the agent's first answer about returns.
- **H4 (brand readiness):** swap Onboarding and ROI, and make required brand facts a go-live check.

8. Day-30 success criteria

These are targets for the 2 pilot brands, measured from the event log. They are product goals, not observed data. If the day-5 gate produces a real baseline, each target is re-set against it.

#	Criterion	Target
1	Pilot brands live on the slice	2
2	Median time to first **meaningful** reply on a new lead	≤ 5 minutes
3	Leads that reach a qualification outcome (qualified / not qualified / not now)	≥ 70%
4	Discovery calls booked through the agent	≥ 15 across both brands
5	Flags claimed within the 10-business-minute SLA (D-06)	≥ 90%
6	Agent answers that contradict the approved facts (weekly audit, 1b)	0

A **meaningful reply** answers the lead's question or asks a qualification question. A plain acknowledgement ("Thanks, we'll be in touch") doesn't count toward criterion 2, even if it's instant.

Also tracked from day 1, without targets until there's a baseline:

- follow-ups completed on time
- duplicate lead records
- missed escalations (from the audit)
- consultant hours per lead, which tests the 1b capacity model

9. How the team uses AI to deliver this

The team uses AI to build faster, in three parallel workstreams that share one data model and integrate daily:

- **Workstream 1:** inbox and CRM UI.
- **Workstream 2:** webhook, data and agent.
- **Workstream 3:** the test set, which is a script of realistic WhatsApp conversations for each pilot brand, used to score the agent's answers.

The aim is a working end-to-end slice inside the first week. Speed of writing code isn't the constraint, though. The things that take real time are Meta approvals, permissions, data quality and the human handoff. Those are proven by the pilot and the audit, not by how fast the code was written.

10. Risks and mitigations

Risk	Mitigation
H1 is wrong: lead handling isn't the main bottleneck	Day-5 gate with pre-agreed switches (§7); days 1–5 do only work that every hypothesis needs
Meta approvals are slow	Apply on day 1; run the pilot on the CRM with human replies until access lands (§6)
The agent gives a wrong answer about money	The agent quotes only approved facts (A32); uncertain or off-script questions get flagged to a human; weekly audit with a 0% threshold
Flags aren't picked up in time	10-business-minute SLA with escalation (1c); criterion 5 tracks it
Pilot brands' lead data is hard to get	Use a scripted test set for the build, and real leads only once the webhook is live
Consultants don't adopt the inbox	Involve them in discovery; the takeover panel replaces their WhatsApp phone, so they don't enter data twice
ROI numbers read as a promise	Assumptions are labelled and editable; benchmarks are shown only when sourced (A33); no guaranteed returns
AI-written code ships defects	Code review, automated tests on the takeover state machine, and the 1c acceptance tests as the release gate
Scope grows past 30 days	Everything outside the §4 thin scope goes to the "later" column; nothing joins it without something else leaving

1b. AI vs Human split: is 76% real?

> **TL;DR.** We mapped all 37 actions across the five modules. Depending on how you count, automation does **68% to 87%** of the work, so "76%" is plausible as a measure of workload. Three things make it misleading as stated:
>
> - **Not all of it is AI.** Of the 87% of time that's automated, AI models do 54% and ordinary software rules (templates, schedules, formulas, logging) do 34%.
> - **It ignores value.** Humans own every one of the highest-value decisions.
> - **It isn't stable.** It swings 14 points depending on how often humans edit the AI's drafts.
>
> The action map makes 76% plausible but doesn't prove it. Our fix is an audited **Autonomy Rate**, computed from logged events and validated in a pilot against a baseline. It's shown as "AI: X% of work | Humans: 100% of closes" (D-05), with the model-vs-rules split on hover, and always paired with outcome metrics.
>
> Beyond the split, each workflow is judged end to end (§7): owner, trigger, inputs, rules, failure path and outcome. Success means a business outcome, not just a task that ran.

1. Action map

All 37 user-facing actions are listed in `ai-human-split.csv`, which imports straight into the Sheet tab `AI_Human_Split`. `ai-human-split.xlsx` is the same data with every formula below already built in, and it's live as a Google Sheet. Each action has an owner:

- **AI:** done end to end by AI.
- **AI→H:** AI drafts, a human approves.
- **H:** human only.

Owner says who is accountable. A second column, **engine**, says what kind of automation does the work, so ordinary software isn't counted as AI:

- **model:** an AI model does it (extraction, drafting, conversation, summaries).
- **rules:** fixed software does it (templates, schedules, formulas, logging, rule-based scores).
- **human:** no automation.

Module	Actions	AI	AI→H	H	Manual hours / brand / month (no AI)	Human hours left with AI
1 Onboarding	7	4	1	2	1.2	0.3
2 ROI Engine	7	4	1	2	2.8	1.0
3 Collateral Studio	6	5	1	0	16.0	1.0
4 Sales & Marketing Agent	10	9	1	0	334.7	2.0
5 HITL CRM	7	3	1	3	102.0	54.6
Total	**37**	**25**	**5**	**7**	**456.6**	**58.8**

Human-only actions, and why they stay human:

- **Onboarding:** uploading the files; the go/no-go decision to take a brand on.
- **ROI Engine:** agreeing the fee, royalty and investment with the brand; presenting the model to the founder.
- **CRM:** taking over a flagged chat; running the discovery call; negotiating and signing.

All four actions with the top value weight (5) are human.

Who does what, by module (from the engine column):

Module	AI models	Software rules	Humans
1 Onboarding	Pre-fill the brand form (founder confirms); extract P&L figures; explain gaps and fixes	Revenue-split chart; Readiness Score v0 (a rule-based rubric, A31)	Upload files; go/no-go on the brand
2 ROI Engine	Draft the Model Card (consultant approves); find comparable benchmarks	3-year projections and sensitivity (formulas); insights v0 (rule-based, per the prototype C plan)	Agree fee, royalty and investment; present to the founder
3 Collateral	Generate deck, brochure, ads and WhatsApp scripts; regenerate; draft the send (human approves)	—	Approve before anything goes out
4 Sales Agent	Qualify on WhatsApp; answer FAQs from approved facts; voice qualification; detect flags; draft ad changes (human approves)	Capture, consent and dedupe; template greeting; email sequence; follow-up nudges; live counts	Approve ad spend changes
5 CRM	Takeover summary; live call notes; suggest deal stage (human confirms)	Interaction timeline	Take over chats; discovery call; negotiate and sign

Counts: 14 actions are AI-model only, 11 are software rules, 5 are AI drafts with human approval (all model-drafted), and 7 are human only.

Inputs are estimates, not data.

- Monthly volumes come from A19: 500 leads, about 150 voice calls, 60 takeovers, 20 discovery calls and 3 signings per brand.
- Standard minutes and the 1–5 value weights come from A20.
- One-off brand set-up actions are spread over a 6-month engagement (A34).

All of these get calibrated in the pilot. The value of this section is the method, not the precise percentages.

2. The split, measured three ways

Method	Formula (Sheet tab `AI_Human_Split`, rows 2–38)	Result
Count, strict (AI only)	`=COUNTIF(D2:D38,"AI")/COUNTA(D2:D38)`	**68%** (25 / 37)
Count, generous (AI→H credited to AI)	`=(COUNTIF(D2:D38,"AI")+COUNTIF(D2:D38,"AI→H"))/COUNTA(D2:D38)`	**81%** (30 / 37)
Time-weighted (share of manual minutes done by AI)	`=SUMPRODUCT(E2:E38,F2:F38,G2:G38)/SUMPRODUCT(E2:E38,F2:F38)`	**87%**
Value-weighted (share of value weight done by AI)	`=SUMPRODUCT(H2:H38,G2:G38)/SUM(H2:H38)`	**68%**
Time-weighted, **AI models only**	`=SUMPRODUCT((J2:J38="model")*E2:E38*F2:F38*G2:G38)/SUMPRODUCT(E2:E38,F2:F38)`	**53.6%**
Time-weighted, **software rules only**	same, with `"rules"`	**33.5%**

Columns: D = owner, E = frequency per month, F = manual minutes, G = AI share of the minutes, H = value weight, J = engine.

By value, the 68% splits into 45.7% models and 22.4% rules. The rules share is mostly high-volume Sales Agent plumbing: the email sequence alone is 50 of the 153 hours that rules take over.

How sensitive the time-weighted number is

Change to the inputs	Time-weighted AI share	Human hours / brand / month
Base case	87.1%	58.8
Agent conversations take half as long to do by hand	80.0%	57.8
Leads halved to 250 a month	79.4%	58.8
Takeovers doubled to 120 a month	83.8%	78.8
Every AI→H approval saves only 50%	86.6%	61.4
Humans edit 20% of the agent's work	**72.8%**	**124.1**
Day-30 slice only (D-04): unshipped AI work done by people	**76.5%**	**89.6**

Two findings matter most:

- **The human edit rate moves the number more than any volume assumption.** That's why the Autonomy Rate below only credits AI for work no human had to edit.
- **The day-30 slice lands at about 76%.** This assumes the agent's work isn't edited and that consultants and designers still do the onboarding, collateral and ad work by hand. The brief's figure is reachable early, but only as a measured outcome, not a constant.

3. Verdict: is 76% AI realistic?

As a claim about…	Verdict	Why
Workload	**Yes**	68–87% depending on method; 76.5% for the day-30 slice
Value	**No, not as stated**	Value-weighted share is 68%. Humans own every top-value decision: go/no-go, terms, discovery call, close.
A live number at launch	**Only if measured**	It's 76.5% at day 30 and 87% at full build, so it changes as modules ship
"AI" specifically	**No, not as worded**	AI models do 54% of the time; 34% is ordinary software that any CRM could do. A badge that says "AI" for all of it overstates the AI.
A hardcoded label	**No**	A 20% edit rate moves it 14 points. A fixed number would be wrong most of the time, and an investor can't audit it.

Testing the brief's own claims

The brief says…	What the action map shows	Holds?
"AI-driven automation (70–80%)"	68–87% depending on the method; 76.5% for the day-30 slice. AI models alone: 54% of time.	**Yes for automation; no for "AI-driven".** The range holds only if templates, schedules and formulas count as AI.
"Human intervention only for high-value actions"	6 of the 7 human-only actions have value weight 4–5. The exception is uploading the deck and P&L (row 2, weight 1), which only the founder can do because only they hold the files. Humans also approve 5 AI drafts, and 2 of those are low-value (brand details form, deal stage, both weight 2).	**Almost.** The upload can shrink (pull files from a shared drive) but not disappear. The 2 low-value approvals should become AI-only once the audit shows the drafts are accurate.
"AI: 76% | Human: 24%"	Human share is 13% of the time, 32% of the value and 32% of the actions (strict count), so "24%" doesn't match any single measure. And 93% of the remaining human hours (54.6 of 58.8) are in the CRM: takeovers, discovery calls and closing.	**No.** "Human: 24%" presents people as overhead to squeeze, when their hours are where the revenue event happens. That's why the badge becomes "Humans: 100% of closes" (D-05).

4. What this means commercially: consultant capacity

Without AI, one brand needs about **457 hours of work a month**. With AI, about **59 human hours** remain, almost all of them in CRM: takeovers, discovery calls and closing. At 160 productive hours per consultant per month (A35):

- **Without AI:** about **0.35 brands per consultant**, roughly 3 people per brand.
- **With AI:** about **2.7 brands per consultant**, about 8× the leverage.
- **If humans edit 20% of the agent's work:** about **1.3 brands per consultant**.

This is the investor story, not the percentage. Leverage is measured in brands per consultant and hours per signed franchisee: about **20 human hours** per signing with AI versus about **152** without (59 h and 457 h ÷ 3 signings).

5. The fix: an audited Autonomy Rate

Definition (A5)

Autonomy Rate = the share of standard minutes of funnel work completed by AI **with no human edit**, per brand, over a rolling 28 days.

```
Autonomy Rate = Σ std_minutes(AI actions, not edited) ÷ Σ std_minutes(all actions)
```

Three numbers, kept apart

The word "automation" covers three different measures. They're never mixed in one figure:

Measure	Formula	Answers	Used for
A. Autonomy Rate (the badge)	AI-completed std minutes, unedited ÷ all std minutes in scope	How much of the whole workload runs without people?	The badge X
B. Success rate on eligible work	AI completions that pass the audit ÷ AI attempts	When AI is allowed to act, how often is it right?	Deciding which actions may run unapproved
C. Human time saved	(baseline human minutes − human minutes now) ÷ baseline	Did people actually get time back?	The capacity story in §4

In this model, A and C come out equal (87%) because the human minutes are derived from the same table. In the pilot they will differ: reviews, rework and corrections add human minutes that A doesn't see. That's why C is measured against a real baseline, not computed from the table.

Unedited-approval rule

- An **AI** action counts as AI.
- An **AI→H** action counts as AI only if the human approves it **unchanged**.
- If the human edits it, rejects it or rewrites it, it counts as human.
- A conversation that gets flagged counts the takeover minutes as human.
- An AI action that fails the weekly audit (wrong answer, missed escalation) counts as human, because someone had to fix it.

Event schema

Every action writes one `ACTION_EVENT`, which matches the platform data model:

Field	Type	Notes
`id`	uuid	
`brand_id`	uuid	Rate is computed per brand, then rolled up
`action_type`	enum (37 values)	Matches `ai-human-split.csv`

Field	Type	Notes
`actor`	`ai_model` | `software` | `human`	Who completed it. Splitting AI models from software keeps the badge honest.
`ai_draft_edited`	bool	True if a human changed an AI draft before it went out
`approved_by`	uuid, nullable	For AI→H actions
`std_minutes`	int	Looked up from a **versioned** standard-minutes table, never typed in by hand
`std_minutes_version`	string	So a past rate can be recomputed after a recalibration
`conversation_id`	uuid, nullable	Links to the audit sample
`ts`	timestamptz	

Audit

- **Weekly sample:** a lead consultant reviews **50 randomly sampled AI-handled conversations** across the live brands against a fixed rubric.
- **Two error rates are published next to the badge:**
  - **Missed-escalation rate:** a conversation should have been flagged and wasn't.
  - **Wrong-answer rate:** the agent said something that isn't in the approved facts.
- **Thresholds:** if missed escalations go above 5% or wrong answers above 0% in a week, the badge shows "under review" until the prompt or rules are fixed. These are proposed product thresholds, to be set with the founder.
- **Recalibration:** standard minutes are recalibrated each quarter from timed samples in the pilot (A20). Each change gets a new version.

Investor view

The badge keeps the brief's format but changes what it means (D-05):

> **AI: X% of work | Humans: 100% of closes**

X is the live Autonomy Rate. Hovering over it shows the definition, the split between AI models and software rules, the audit date and the two error rates. Next to it go three outcome metrics, so autonomy is never shown alone:

1. **Human hours per signed franchisee:** modelled at about 20 h, versus about 152 h without AI.
2. **Cost per signed franchisee:** ad spend + AI cost + human hours × loaded hourly cost, divided by signings. The inputs come from the pilot; no number is claimed here.
3. **Speed to lead:** median minutes to first meaningful reply (not a plain acknowledgement). The target is ≤ 5 (1a).

6. Validating 76%: the pilot plan

76% is treated as a hypothesis to test, not a capability to claim. The pilot is the 2 brands in the day-30 slice (1a §7).

1. **Baseline first (days 1–5).** For each in-scope action: how many instances a month, how long a person takes now, error and rework rates, and what the brand's existing tools already automate. Existing automation is counted as the baseline, never as new AI.
2. **Log every instance.** Every action writes an `ACTION_EVENT`: attempted, completed, escalated, failed, edited.
3. **Count only successes.** An AI action counts only if it's unedited and passes the audit (see the rule above). A fast reply with a wrong fact is a failure, not automation.

4. **Report with the fine print.** Each published rate states the period, the brands, which actions were in scope, what was excluded, and the audit results. Model and rules shares are always shown separately.
5. **Expand by evidence.** An action moves from AI→H to AI only when its success rate (measure B) stays above the agreed bar for 4 consecutive weekly audits. The bar is a proposal to agree with the founder.

Guardrails during the pilot:

- wrong money or commercial claims (target 0)
- missed escalations (≤ 5%)
- duplicate leads
- messages sent without consent

Any breach pauses the badge ("under review").

7. Beyond AI vs human: is the whole workflow dependable?

Labelling each task AI or human is only the first cut. The OS is only useful to franchise teams if the **whole workflow** around each task works, including the parts that aren't AI at all.

What gets evaluated for each workflow

Dimension	What we check	Where this plan already covers it
Deterministic automation	Which steps are fixed rules rather than AI	The engine column (§1); rules do 33.5% of the time (§2)
Data quality	Inputs are complete and correct: lead details, brand facts, P&L	Dedupe by phone number (1a §4); the agent quotes only approved facts (A32)
System integrations	Each outside dependency, and what happens when it's down	WhatsApp access and templates (1a §6); send failures and the agent's Error status (1c §4)
Human approvals	Who approves what, and how fast	The AI→H owner code; the 10-business-minute claim SLA (D-06)
Exception handling	A defined failure path for every step	1c's error handling (including failed and repeated takeovers) and 10 edge cases
Security	Who can see and act on each brand's data	Row-level security per brand (1c §4)
Auditability	Every action can be traced afterwards	`ACTION_EVENT`, `STATE_EVENT` and the weekly audit (§5)
User adoption	Consultants actually use it	The takeover panel replaces their WhatsApp phone (1a §10); flags claimed within the SLA
Scalability	It holds as brands and leads grow	Brands per consultant (§4); sensitivity to volume (§2)
Operating cost	Ad, AI and human cost per result	Cost per signed franchisee (§5, investor view)

Every workflow gets the same six-part definition

Owner · trigger · required inputs · execution rules · failure path · measurable outcome. Example, using the first workflow in the day-30 slice:

Part	New lead → first reply
Owner	The brand's sales manager
Trigger	A Lead Ads form is submitted
Required inputs	Phone number, consent, the brand's approved facts, an approved greeting template
Execution rules	Dedupe by phone number → check consent → send the template greeting → qualify → answer only from approved facts → flag to a human when a HITL rule fires

Part	New lead → first reply
Failure path	Send fails: retry up to 3 times, then flag "Delivery problem". WhatsApp not yet approved: consultants reply through the CRM. Flag unclaimed: escalate to the manager at 10 business minutes.
Measurable outcome	Median time to first **meaningful** reply ≤ 5 minutes; ≥ 70% of leads reach a qualification outcome (1a §8)

A task that ran is not an outcome

The task ran…	…but that doesn't prove	So we measure instead
An automated acknowledgement was sent to a new lead	The lead got a meaningful answer or moved toward signing	Time to first meaningful reply; qualification outcome reached; discovery call booked
A financial projection was generated	Its assumptions are right, or the deal is commercially viable	A consultant approved the Model Card before the agent may quote it (A32); 0 wrong money answers in the audit
A takeover flag was raised	A human actually responded	Flag claimed within the SLA; time from flag to the consultant's first reply

Five lenses for judging automation

Lens	Question	Measure
Successful task completion	Did the step finish correctly, not just run?	B. Success rate on eligible work (§5)
Human effort saved	Did people actually get time back?	C. Human time saved, against the pilot baseline (§5)
Quality	Was the output right?	Wrong-answer and missed-escalation rates from the weekly audit
Reliability	Does it keep working, including when something fails?	Delivery failures and agent uptime; flags claimed within the SLA; 0 AI messages after a takeover (1c §5)
Business impact	Did it move franchisees toward signing?	Speed to lead; discovery calls booked; human hours and cost per signed franchisee

The Autonomy Rate (measure A) is only ever reported next to these five. The goal is an OS that franchise teams can depend on, not just one that's highly automated.

1c. Developer-ready spec: Conversation Takeover (Human-in-the-Loop)

> **TL;DR.** The brief covers human takeover in one line: "Includes Human-in-the-Loop (HITL) takeover". Behind that line sit the hardest questions in the product. Who is allowed to reply in a live WhatsApp chat? What happens when two people click at once? What if nobody picks it up? What if the AI was just about to send something? This spec answers them in plain terms: what the screen shows, how a conversation moves between the AI and a person, what happens when something fails, and how we know it works. Testable numbers: a 10-business-minute pick-up time (D-06), 11 pass/fail tests and pilot metrics. The steps it records also feed the Autonomy Rate (1b).

How we chose this feature

The brief describes several features in one line or less. We asked three questions of each:

1. Is it in the first 30 days (1a)?
2. How much does it cost if a developer guesses wrong?
3. Is the unclear part **what to measure** (which can be tuned later) or **how it behaves** (which has to be right on day one)?

Feature (as the brief puts it)	What's unclear	In the first 30 days?	Cost of a wrong guess	Unclear part
HITL takeover	Who owns the chat; two people clicking at once; nobody picking it up; WhatsApp's 24-hour rule; "stop messaging me"	**Yes**, rank 1	**High**: a lost hot lead, or the AI and a person both replying	**Behaviour**
AI Readiness Score	What is scored, the weights, the thresholds	No	Medium: a scoring rubric can be adjusted (A31)	Measurement
AI insights	What triggers an insight	No	Medium: misleading advice	Measurement
Live call intelligence	What it captures and suggests, and when	No	Medium	Behaviour
Meta ads optimisation	What the AI may change without approval	No	High (ad spend), but outside the 30 days	Behaviour
Live tracking	Which numbers to show	Yes (basic counts)	Low	Measurement

Only takeover is in the first 30 days, expensive to get wrong **and** unclear in behaviour. The close runner-up was the Readiness Score (A18): *takeover is harder to build; the Readiness Score is harder to define.* The 1b numbers back this up: 93% of the human hours left after AI (54.6 of 58.8 a month per brand) sit in the CRM, starting with takeovers.

Platform facts (WhatsApp's 24-hour rule, error 131047, the 4,096-character limit) were checked against Meta's documentation; the sources are at the end. The pick-up time and business hours are our product choices, labelled as such (D-06, A22).

1. Feature overview

Module	AI Sales & Marketing Agent (the WhatsApp agent in the 30-day release)
Feature	Human-in-the-Loop takeover of a live WhatsApp conversation

Objective	Let a consultant step into any AI conversation safely, at the right moment, without the prospect ever getting two replies, and hand it back to the AI when they're done.
Problem statement	The AI handles most conversations, but some need a person: a hot lead, a money question, a complaint, or a prospect who asks for a human. Today nothing defines who replies, how fast, or what happens if the AI and a person both act. That risks lost leads, double replies, and messages sent to people who said stop.
Proposed solution	A clear owner for every conversation at all times (AI, nobody yet, or one named consultant). The AI flags conversations that need a person and explains why. A consultant claims it, and the AI goes quiet. If nobody claims it within 10 business minutes, the manager is alerted. Every step is logged. Separately, the agent itself shows a simple status (Active / Paused / Error), so a team can see whether the AI is really sending, not just switched on.

Who uses it

As a...	I want to...	So that...
Consultant	see which conversations need me, why, and how long I have	I reach hot prospects before they go cold
Consultant	claim a conversation and know nobody else (AI or person) is replying	the prospect never gets two answers
Consultant	hand a conversation back to the AI with a reason	the AI keeps nurturing, and we learn why it flagged
Sales manager	be alerted when a flag sits unclaimed too long, and see if the agent is failing	no hot lead is dropped and no outage goes unnoticed
Prospect	get a person quickly when I ask, and stop all messages when I say STOP	I trust the brand

In the first version: WhatsApp only; one brand per conversation; one consultant owns a conversation at a time; English, Hindi and Hinglish keyword lists.

Later (after day 30): voice and email takeover; a full agent switchboard covering Voice, Email and Meta Ads; automatic round-robin assignment; routing by consultant skill.

2. UI behaviour

A. Agent status (top of the inbox)

The **agent's** status is kept separate from each **conversation's** status. "The AI is switched on" and "the AI is actually sending" are not the same thing, and a team needs to see both.

Agent status	What it means	What the screen shows
Active	The WhatsApp agent is on and messages are going out	Green dot, "Active", and the time of the last successful send
Paused	A manager has paused the agent for this brand (or WhatsApp access isn't approved yet)	Grey dot, "Paused by Ravi at 10:14". Every new message goes to a person (see edge case 9).
Error	The agent is switched on but messages are **not** going out	Red dot, "Error: messages not sending since 10:20", with the last error in plain words and a **Retry check** button

The status strip shows "Enabled" and "Last successful send" side by side, so "on but failing" can't hide. Only a manager can pause or resume the agent.

B. Inbox and live conversation

- **Each inbox row shows:** prospect name or number, brand tag, last message, and a coloured status chip: AI (blue `#2563EB`), Needs a human / Escalated (amber `#F59E0B`), Human (green `#10B981`), Closed / Opted out

(grey).

- **"Needs human" filter:** shows conversations waiting for a person, most urgent first. A countdown shows the time left to pick it up; it turns red under 2 minutes.
- **New flags appear** on every consultant's screen within 2 seconds, without refreshing.
- **The thread** shows prospect, AI and human messages, plus system notes in line: "Flagged: high intent", "Claimed by Priya", "AI reply cancelled: Priya took over", "Returned to AI: question answered".

C. Human takeover

Buttons in the conversation header (the header also shows the assignee and how long is left in WhatsApp's 24-hour reply window):

Conversation status	Buttons shown
AI handling	**Take over**
Needs a human	**Claim**
Escalated	**Claim** (managers also see **Assign**)
Human handling	**Return to AI**, **Mark Won**, **Mark Lost**
Won / Lost / Opted out	None (read-only)

"Why flagged" card. Pinned at the top while the conversation waits for a person. It lists every reason in plain words with the actual value, e.g. *Deal value: stated investment ₹45L, brand threshold ₹40L*; *Asked for a person: "call me"*.

Takeover panel (right side, once a consultant owns the chat): a 5-point AI summary (budget, city, timeline, objections, last unanswered question), the lead score, and one suggested reply. The suggestion goes into the message box only when clicked and is **never sent automatically**.

Message box

Situation	Message box
AI handling	Locked: "AI is handling this conversation. Take over to reply."
Human handling	Open, with an "AI paused" banner and a character counter (`n / 4,096`)
More than 24 h since the prospect's last message	Free typing is locked; a picker of approved WhatsApp templates appears
Won / Lost / Opted out	Hidden; read-only thread

3. State transitions

Conversation states (system code in brackets, for developers):

State	Who may send	Meaning
AI handling (`AI_ACTIVE`)	AI only	The default. The AI qualifies and answers from approved facts.
Needs a human (`FLAGGED`)	Nobody (the AI has sent one holding message)	A takeover rule fired. Waiting for a consultant. The 10-minute clock is running.
Escalated (`ESCALATED`)	Nobody	Nobody claimed it in time. The manager has been alerted.
Human handling (`HUMAN_ACTIVE`)	The assigned consultant only	The AI is silent in this conversation.
Won / Lost (`CLOSED_WON` / `CLOSED_LOST`)	Nobody (final)	Deal closed.

State	Who may send	Meaning
Opted out (`OPTED_OUT`)	Nobody, not even templates (final)	The prospect asked us to stop.

How a conversation moves

#	Current state	Trigger	Next state	Expected behaviour
T1	AI handling	A prospect message matches at least one takeover rule (Task 3)	Needs a human	The flag lists every rule that fired and why. The AI sends **one** holding message and nothing else. The 10-business-minute clock starts. On-duty consultants are notified.
T2	AI handling	A consultant clicks **Take over** (no flag needed)	Human handling	The conversation is assigned to them; the AI goes silent; **any AI reply already queued is cancelled**; a note "AI paused" appears in the thread.
T3	Needs a human	A consultant clicks **Claim**	Human handling	First click wins. The clock stops and the claim time is recorded. Any queued AI reply is cancelled.
T4	Needs a human	10 business minutes pass with no claim	Escalated	The brand's manager gets an in-app and email alert; the conversation goes to the top of their queue.
T5	Escalated	The manager assigns a consultant or claims it	Human handling	Assigned; the time it spent escalated is recorded.
T6	Human handling	The consultant clicks **Return to AI**	AI handling	Only allowed with a reason, every legal or money flag marked resolved, and an empty message box. The AI replies again from the next prospect message.
T7	Human handling	The prospect's last message has gone unanswered for 30 business minutes	Needs a human	Re-flagged as "No human reply" with a new clock. The original consultant gets the first 2 minutes to claim it back, then anyone can.
T8	Human handling	The consultant marks the deal **Won** or **Lost**	Won / Lost	The deal record closes and the message box locks. Lost needs a reason.
T9	Any open state	The prospect sends "STOP", "unsubscribe", "band karo" or "message mat karo" (any capitals; the whole message)	Opted out	One confirmation is sent, the only message allowed. All sending is blocked, templates included. Consent is recorded as withdrawn. Any clock is cancelled.

Behind every change the system adds 1 to the conversation's version number (`state_version`) and writes a log row: who, from, to, why, when. Every button sends the version it saw; if someone else changed the conversation first, the server refuses (HTTP 409) and nothing changes. This is how "first click wins" and "no double replies" are enforced.

Agent status changes (separate from the conversation states above)

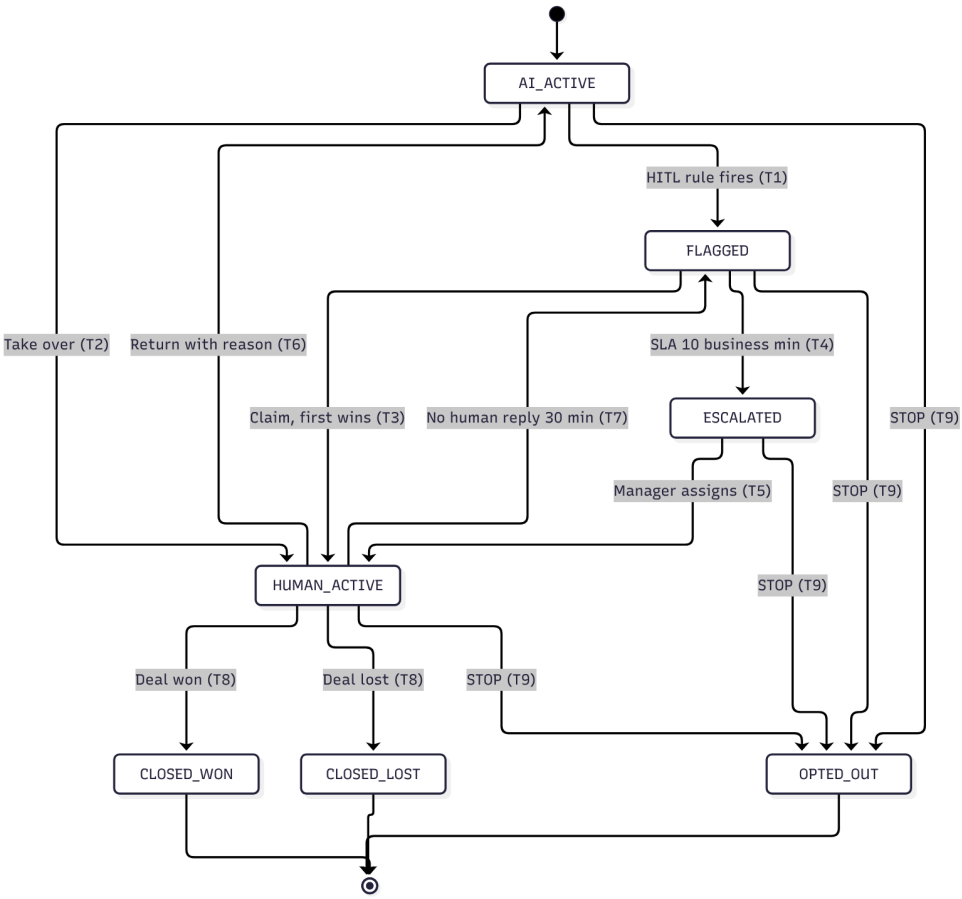
Current	Trigger	Next	Expected behaviour
Active	A manager clicks **Pause agent**	Paused	No AI messages for this brand; new prospect messages go to "Needs a human" (edge case 9)
Paused	A manager clicks **Resume**	Active	The AI handles new messages; conversations already with a person stay with them
Active	Sends keep failing (proposed: 3 in a row), or the WhatsApp provider reports the connection is down	Error	Red status; the manager is alerted; new prospect messages go to "Needs a human" so no lead waits on a broken AI
Error	A send or health check succeeds	Active	Status turns green; the outage start and end times are logged

Business hours: 09:00–21:00 IST, every day, in the first version. This can be changed per brand and is our product choice (A22, D-06). A flag raised outside those hours starts its clock at 09:00.

Holding message (English shown; Hindi and Hinglish versions are also stored):

> "Thanks, a member of our franchise team will reply to you here shortly." Outside hours, "shortly" becomes "after 9 am".

The holding message is only an acknowledgement, so it does **not** count as the first meaningful reply in the speed-to-lead target (1a §8). That clock stops at the consultant's first real reply.



State diagram: the 7 states and transitions T1–T9 from the table above.

4. Error handling

Scenario	Expected behaviour
An AI reply is queued when a consultant takes over	Just before any AI message is sent, the system re-checks that the conversation is still "AI handling". If not, the message is cancelled, never sent, and logged as "AI reply cancelled: human took over".
Two consultants claim at the same moment	The first click wins. The second sees "Already claimed by Priya at 14:02" and their screen updates to show Priya as owner. Nothing else changes.
The same consultant clicks Take over or Claim twice (double-click, slow network)	Treated as one request: one assignment, one log entry, no error shown.
Takeover fails (network or server error)	The conversation stays in its previous state, so the AI is not silenced with nobody in charge. The consultant sees "Couldn't take over. Try again." A retry is safe and can't create a second assignment.
Return to AI fails	The conversation stays with the consultant, who sees "Couldn't hand back. You still own this conversation." A retry is safe.

Scenario	Expected behaviour
A message fails to send (WhatsApp provider error, or no answer within 10 s)	The bubble turns red, "Not delivered", with **Retry**. The text is kept. A retry can't send the message twice. After 3 failed retries the conversation is flagged "Delivery problem".
The WhatsApp connection is down	The agent status turns **Error** and the manager is alerted. Failed sends are logged as failures and **never counted as AI work done** (they don't raise the Autonomy Rate). New prospect messages go to a person.
More than 24 hours since the prospect's last message	Free typing is replaced by the template picker: "More than 24 h since the prospect's last message. Send an approved template." If WhatsApp still returns error 131047, the bubble says "Window closed. Send a template." Free text is never retried automatically.
The live connection drops	Banner: "Reconnecting… messages may be out of date". **Send** is disabled until the screen is up to date again, so nobody replies to an old view.
The prospect writes again while waiting for a person	The new message shows in the thread; the "Why flagged" card updates if more rules fire. No second holding message; the clock does not restart.
Nothing needs a person	"No conversations need you right now", plus how many the AI is handling.
Screens loading	Grey placeholder rows and bubbles, not a full-page spinner. The last 50 messages load first; older ones load on scroll.

Edge cases

1. **A flag fires at 20:58.** About 2 business minutes count today; the clock resumes at 09:00. The holding message says "after 9 am".
2. **Several rules fire on one message.** One flag lists every reason. There is never more than one flag per message.
3. **A closed lead messages again.** A new cycle starts in "AI handling", with the full history kept and the previous outcome shown in the header.
4. **STOP during a human conversation.** T9 applies: the message box locks and the consultant sees "Prospect opted out". Only the prospect sending "START" can re-open it, which records new consent.
5. **The consultant goes offline mid-conversation.** If the prospect's message sits unanswered for 30 business minutes, T7 re-flags it.
6. **The 24-hour window closes while the consultant is typing.** The draft is kept; Send becomes "Send as template"; nothing goes out as free text.
7. **The assigned consultant's account is deactivated.** Their conversations are re-flagged at once as "Assignee unavailable".
8. **A consultant takes over just as the AI's reply is about to go.** The pre-send check cancels the AI reply (see above). The prospect sees only the consultant's message.
9. **The AI is paused or in Error for a brand.** This covers the period before WhatsApp approval arrives (the fallback in 1a §6), a manager pausing the agent, and an outage. Every new prospect message goes straight to "Needs a human" with reason "AI off for brand". The same 10-minute rule applies; the AI sends nothing except the holding message, and in Error not even that.
10. **The same prospect arrives again as a new lead.** A Lead Ads form whose phone number matches an open conversation for the same brand is added to that conversation's timeline. It doesn't create a second conversation or change its state.

Validation rules

Field or action	Rule
Message text	1–4,096 characters after trimming; blank messages are rejected. 4,096 is WhatsApp's limit (Meta text messages).
Any button that changes the state	Must carry the version the user saw; if it's out of date the server refuses (HTTP 409)

Field or action	Rule
Take over / Claim	Must carry a request ID, so a repeated click is treated as the same request
Return to AI	A reason is required: *Question answered* / *Not ready, keep nurturing* / *Wrong flag* / *Other* (5–200 characters). Every legal or money flag must be marked resolved first.
Mark Lost	A loss reason from a fixed list
Template send	Every template field filled; only templates WhatsApp has **approved** are listed
Any send	Blocked if the prospect opted out, the conversation is closed, or (for AI sends) it is no longer "AI handling"
Who can act	Only consultants or managers of that brand can see, claim or take over its conversations (row-level security); only managers can pause the agent

5. Success criteria

Functional acceptance (pass/fail tests)

#	Situation	Passes when
AC1	AI handling; a prospect message matches at least one rule	"Needs a human" within **2 s** (95% of the time). The card lists **every** rule that fired, with its value. Exactly **1** holding message, then **0** AI messages.
AC2	Two consultants claim the same flag within **500 ms**	Exactly **1** succeeds. The other sees the "already claimed" notice within **1 s**. The version number goes up by exactly **1**.
AC3	A flag nobody claims	Escalated at **10 business minutes** (± 30 s) and the manager is alerted. A flag raised outside hours escalates at **09:10** at the earliest.
AC4	Human handling; Return to AI with no reason, or with an unresolved legal/money flag	The button stays disabled. **0** state changes.
AC5	Message box holds **4,097** characters	Send is disabled and the counter is red. At **4,096** it sends.
AC6	More than 24 h since the prospect's last message	Free typing is locked and the template picker shows. An error 131047 marks the bubble failed, with **0** automatic free-text retries.
AC7	"STOP" or "band karo" in any open state	Opted out within **2 s**. **0** messages afterwards, templates included. Consent is recorded as withdrawn.
AC8	Human handling; the prospect's message goes unanswered	Re-flagged "No human reply" at **30 business minutes** (± 30 s).
AC9	An AI reply is queued, then a consultant takes over before it sends	**0** AI messages reach the prospect after the takeover. The cancelled reply is logged.
AC10	The same Take over / Claim is sent twice, or a takeover or Return to AI fails	Exactly **1** assignment and **1** log entry for a repeat. After a failure the conversation is in its previous state, and a retry succeeds without duplicates.
AC11	Sends to WhatsApp fail (simulated outage)	Agent status shows **Error** within **60 s** (proposed); the manager is alerted; the failed sends are logged as failures, with **0** counted as completed AI work.

Prototype B's test script runs AC1, AC2, AC4, AC5, AC7, AC9 and AC10 directly. AC3 and AC8 use a shortened timer in the demo; AC6 and AC11 are simulated.

Pilot metrics (after launch)

Targets are proposals, to be set properly once the 2-brand pilot has a baseline.

Metric	Proposed target	Where it comes from
Takeover success rate: takeovers that ended with the consultant in charge and the AI silent, out of all attempts	≥ 99%	Conversation log
AI messages after a takeover (the AI sent something after a person took charge)	0	Message log
Flags claimed within 10 business minutes	≥ 90% in month 1, ≥ 95% by month 3	Flag and claim times
Median time to claim	≤ 5 business minutes	Conversation log
Median time from flag to the consultant's first real reply	≤ 10 business minutes	Message log
Integration reliability: share of the time the agent is Active while switched on, and sends delivered on the first try	Baseline in month 1	Agent status log, send log
Human handling time vs baseline: consultant minutes per takeover, compared with handling the same chat by phone before the pilot	Baseline in days 1–5 (1b §6); goal: lower	Conversation log, baseline timings
Missed escalations found in the weekly 50-conversation audit (1b)	≤ 5%	Audit sheet
"Wrong flag" share of Return-to-AI reasons	Tracked; above 30% triggers a review of the rules	Flag reasons
Takeovers that lead to a booked discovery call	Baseline in month 1; no target until we have data	Deal stage

6. Definition of done

The feature is done when:

- ☐ All 11 acceptance tests pass on a scripted test set.
- ☐ Automated tests cover the risky paths: a failed takeover, a failed Return to AI, a queued AI reply at takeover, a repeated request, and the AI and a person acting at the same moment.
- ☐ Every state change, cancelled AI reply and failed send writes a log entry, and the events below reach the Autonomy Rate.
- ☐ The agent status (Active / Paused / Error) is visible, and the manager alert has been tested with a simulated outage.
- ☐ Holding and opt-out messages exist in English, Hindi and Hinglish, and the consultants who will use it have reviewed them.
- ☐ Only that brand's consultants and managers can see or act on its conversations (checked with a user from another brand).
- ☐ **Controlled rollout:** switched on for the 2 pilot brands only (A21), watched daily for the first week against the pilot metrics, with a manager able to pause the agent at any time. Wider rollout only after 2 weeks with 0 AI messages after a takeover and 0 messages to opted-out prospects.

Events this feature writes (for the Autonomy Rate)

Each step writes an `ACTION_EVENT` (1b §5). This is how the steps count toward the badge:

Step	Action map row	Done by (`actor`)	Counts as
Detect the rule and raise the flag (T1)	29 Detect HITL triggers	`ai_model`	AI. It becomes human if the weekly audit finds a missed escalation.
Send the holding message	23-style template	`software`	Automation, shown as software rules, not AI

Step	Action map row	Done by (`actor`)	Counts as
Build the takeover summary	32 Summarise lead	`ai_model`	AI
Consultant replies, including a suggested reply sent unedited	33 Take over and reply	`human`	Human. A takeover is human work by definition (1b's unedited rule applies to AI-drafted actions a human approves, not to takeovers).
Mark Won / Lost (T8)	37 Update deal stage	`human` (confirms an AI suggestion)	AI only if the suggested stage is accepted unchanged
An AI send that failed or was cancelled	—	`ai_model` or `software`	Logged as failed or cancelled. **Never** counted as completed work.

Platform facts used (verified)

- **24-hour window and error 131047:** "More than 24 hours have passed since the recipient last replied to the sender number. Send the recipient a template message instead." (Meta error codes, checked 15 Oct 2026).
- **Text message limit:** 4,096 characters (Meta text messages; also stated in 360dialog's docs).

Appendix: assumptions and decisions

Every assumption (A) and decision (D) cited in this document. Assumptions are stated openly so they can be challenged; none of the inputs are presented as measured data.

Assumptions

ID	Topic	Assumption	Basis
A1	Business model	Scale100x earns a one-time setup fee, a monthly retainer, and a success fee per franchisee signed. Values are editable inputs.	Standard consultancy structure. Scale100x's pricing is **not public** (only job postings found).
A5	76% metric	Undefined in the brief. Computed three ways. Proposed **Autonomy Rate**: time-weighted share of funnel work completed by AI with no human edit.	Brief leaves it open
A7	Capacity	2 full-stack engineers + 1 designer + PM, using AI tools, building on React + Supabase from close to scratch	Brief gives no team size
A18	1c feature	Conversation Takeover chosen over the AI Readiness Score	Takeover is harder to build; Readiness is harder to define. Readiness is still prototyped in A (A31).
A19	Volumes	Per brand per month: 500 leads, about 150 voice calls, 60 takeovers, 20 discovery calls, 3 signed	Illustrative and internally consistent, **not sourced**
A20	Weights	Standard minutes and 1–5 value weights per action are candidate estimates, to be calibrated in the pilot	Estimate
A21	Pilot	30-day release goes to 2 existing brands	Scale100x already has 20+ onboarded brands
A22	SLA	Flag must be claimed within 10 business minutes; business hours 09:00–21:00 IST	Product choice, not research (D-06, confirmed)
A31	Readiness Score	Rule-based v0 rubric: 5 weighted sub-scores (unit economics, proof of concept, documentation, channel concentration, franchise terms), each Red/Amber/Green with the rule and threshold shown. Thresholds are design choices, to be calibrated on real outcomes.	The brief doesn't define the score
A32	Single source of numbers	Prototype C uses the Task 5 financial model's base-case inputs and formulas, so the same inputs give the same outputs. Prototype B's agent quotes only C's approved Model Card facts.	Avoids three different FreshBrew numbers
A33	Benchmarks	India + US benchmarks are shown only with a source and date; anything unsourced is labelled "illustrative" or left out	No benchmark data in the brief; no invented numbers
A34	Set-up amortisation	One-off brand set-up actions (onboarding, model, collateral) are spread over a 6-month engagement: frequency 1/6 per month in the action map	Estimate; makes per-brand-per-month hours comparable
A35	Consultant capacity	One consultant has 160 productive hours a month (8 h × 20 working days)	Estimate

Decisions

ID	Decision	Status	Date
D-04	30-day order: Sales Agent + CRM slice → thin ROI → thin Onboarding → Collateral deferred	Confirmed	15 Oct 2026
D-05	Badge reframed as "AI: X% of work | Humans: 100% of closes", where X is the live, audited Autonomy Rate (1b)	Confirmed	15 Oct 2026
D-06	Takeover SLA of 10 business minutes, 09:00–21:00 IST	Confirmed	15 Oct 2026